Bitwise Operators

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1. Write a prorgram for the following one.
   a) Set a bit b) Clear a bit c) Toggle a bit

2. WAP to find the given number is even or odd using bitwise operators.

3. WAP to find the given number is +ve or -ve using bitwise operators.

4. WAP to swap two numbers using bitwise operators.

5. WAP to find the given number is power of 2 or not.

6. WAP to find the given number is divisble by 8 or not using bitwise operators.

7. Write a program to rotate the bits. Input the no.of rotations, at runtime.
   Ex : binary : 10000000000000000000000000001011
   rotations : suppose 3 times right, then
   result : 01110000000000000000000000000001

   binary : 10000000000000000000000000001011
   rotations : suppose 4 times left, then
   result : 00000000000000000000000010111000

8. Convert the characters Upper to Lower and Lower to Upper using bitwise operators.

9. Write a program to reverse the bits of a given number.
   Note : not just reverse printing.

10. Write a one line code to compare two numbers equal or not using bitwise operators.

11. Write a program to print float binay formation using char *ptr.

12. Write a program to swap the adjucent bytes of a given 4-digit hexadecimal number.
    Ex : given number = 0x1234;
    after swap : 0x3412;

13. Write a program to delete no.of bits from particular position in a given number.
    Input the no.of bits, at runtime.
    Ex: Suppose num = 100;
    It's Binaray is 00000000000000000000000001100100
    delete 2 bits from 4th position
    then result is 00000000000000000000000000011100

14. Write a programme for swapping first and last nibbles in a given integer.
    Ex: Suppose num = 10

It's Binary is 00000000000000000000000000001010
After swap 10100000000000000000000000000000

15. Write a logic to extract P bits from Posion N in an integer M

16. Write a macro to clear a bit at the position N in an integer M.

17. There are 48 bits are stored in an array of character buffer and store them into 2 integer variables.

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